

# Strategy-Decision Memo: Next Research Lane for Coinbase RTB Bot

## Executive verdict

**Primary lane (best next move right now): Lane A — OHLCV-only, spot-only, multi-asset rule library / portfolio / ensemble research.**

**Backup lane: Lane D — BTC-only indicator mean reversion fallback (as a baseline + fast falsification path, not as the main bet).**

This is **not** because “candles are easy.” Lane A is first because it is the only lane that (1) can be made research-credible without abandoning spot, (2) directly fixes the structurally broken part of the prior process (single-symbol BTC search), and (3) has *some* supportive crypto-specific evidence for diversified / portfolio-based predictability while also acknowledging mixed BTC out-of-sample findings. <sup>1</sup>

Lane ranking for *current project state* (constraints + 2-week falsifiability):

**Lane A > Lane D > Lane B > Lane C.**

## Capability-to-edge matrix

### Binding repo constraints that actually constrain lane choice

Only the facts that bind strategy choice (and where they come from):

- **Spot-oriented, paper-first, single-process execution scaffold.** Paper mode is explicit in config (`runtime.mode: paper`) and the symbol list is currently a single spot pair (`BTC-USD`). <sup>2</sup> The Analytical Review PDF explicitly characterizes the runtime as “single-process... paper-mode orchestration.” (Analytical Review PDF, p.1 — extracted text shows “single-process... paper-mode orchestration.”)
- **Breakout/retest logic is still hard-coded even though archived.** The state machine literally encodes breakout → retest → continuation logic with fixed thresholds (e.g., breakout above 20-bar high, volume multiple, RSI band, close percentile, retest window, ATR gates) and emits `LONG` only. The `research_plan.md` marks the breakout-retest-continuation strategy as “formally archived” (`research_plan.md` “Final Conclusions” section).
- **Config and runtime risk/slippage are misaligned.** `config.yaml` sets `risk.risk_per_trade: 0.0035` and `execution.max_entry_slippage_bps: 10` while `bot/risk.py` hard-codes `MAX_RISK_PERCENT = 0.002` and `MAX_SLIPPAGE = 0.005` (50 bps).

- **No research-grade backtest / walk-forward / parameter-sweep / multiple-testing subsystem exists.** The Analytical Review PDF explicitly calls out the need for a “credible backtest + walk-forward + parameter sensitivity + multiple-testing defense” and sketches missing components (walk-forward skeleton, multiple-testing module). (Analytical Review PDF, pp.1, 3, 10–12) This is consistent with the repo still embedding one strategy into runtime logic rather than a research harness. <sup>3</sup>

A critical “quiet constraint” that matters for falsification cost: **paper fills assume zero commission** (PaperAdapter returns `commission: 0.0`). <sup>4</sup> This matters because any lane claiming OHLCV edge is extremely sensitive to realistic fees + slippage (see Anghel 2022 on false discoveries when costs are omitted). <sup>5</sup>

## Capability-to-edge matrix (A vs B vs C vs D)

A OHLCV spot multi-asset portfolios

**Breadth + selection discipline + portfolio construction:**

test a *pre-registered* multi-asset rule library, apply multiple-testing control, then allocate as an ensemble/ portfolio so edge doesn't depend on one BTC rule.

Large-scale crypto technical-rule study finds profitability after multiple-hypothesis procedures and explicitly reports **BTC has no OOS predictability**

while other cryptos do — supports “multi-asset, not BTC-only.” <sup>6</sup> A separate Bitcoin-focused study uses 75,360 rules, applies multiple-hypothesis selection, and forms portfolios; reports **OOS outperformance vs buy-and-hold** risk–return wise — supports “portfolio-of-rules” framing (still not guaranteed durable). <sup>7</sup> Cross-crypto return predictability literature supports cross-sectional / portfolio approaches and OOS long-short profitability after costs. <sup>8</sup>

Data-snooping is endemic in single-history time series; White (2000) exists specifically because repeated reuse produces false winners. <sup>9</sup> Even with FDR, persistence and costs can wipe out “winners” in classic settings; selection ex ante can fail. <sup>10</sup> Costs omission inflates false discoveries even under modern multiple-testing procedures. <sup>5</sup>

OHLCV per symbol (timestamp-open, open/high/low/close/volume). Prefer 1h/4h/1d for alignment with existing timeframes. Coinbase candle endpoint supports `ONE_MINUTE` through `ONE_DAY`, max 350 candles per request → requires pagination. <sup>11</sup>

Spot pairs only. Needs ability to fetch candles for many spot products and execute spot orders (eventually).

High fit market (spot) b  
**fit on research workflow**  
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| Lane                                         | Mechanism<br>(edge hypothesis)                                                                                                                                               | Best supporting evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Strongest evidence against it                                                                                                                                                                                                                                                                                                                                                    | Required data (exact)                                                                                                                                                                                                                               | Required market/instrument access (exact)                                                                                                                                                                                                                         | Fit with current                                                                                           |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>B derivatives carry / basis / funding</b> | <b>Structural carry:</b> cash-and-carry basis capture and/or funding-rate capture driven by segmentation, leverage demand, and limits-to-arbitrage (not indicator patterns). | <p>BIS “Crypto carry” documents large basis, average &gt;10% annually, sometimes &gt;40%, and finds <b>high carry predicts crashes</b> (edge exists but is tied to tail risk). <sup>12</sup></p> <p>Perpetual futures theory work explains funding-based anchoring and shows deviations comove/diminish; implies potential arbitrage Sharpe but not free. <sup>13</sup></p> <p>Coinbase docs confirm perps exist via Advanced Trade for eligible regions with margin requirements and liquidation thresholds → implementable in principle. <sup>14</sup></p> | <p>Same BIS evidence: carry rises in booms and predicts crashes; limits-to-arbitrage include regulatory + margin frictions → you’re paid for bearing crash/liquidation risk, not harvesting a stable yield. <sup>12</sup></p> <p>Coinbase liquidation docs explicitly warn losses can exceed margin/collateral and liquidation can be worse-priced/higher fee. <sup>15</sup></p> | <p>Funding rates (per hour), perp mark price, index/spot mark price, basis history, fee schedules, margin parameters, liquidation events. Coinbase discloses funding calc mechanics (VWAP/TWAP sampling every 3 minutes, scaled). <sup>16</sup></p> | <p>Requires derivatives account eligibility + onboarding; margin collateral (USDC) and ability to trade perps; possibly separate regulated entity depending on region. Coinbase perps access is gated by “eligible regions” + onboarding steps. <sup>17</sup></p> | <b>Low fit:</b> baseline spot + p risk eng not mar aware. Y would b adding entire liquidat margin sub syst |

| Lane                                             | Mechanism<br>(edge hypothesis)                                                                                                                                          | Best supporting evidence                                                                                                                                                                                                                                                                                                                                                                                                                                    | Strongest evidence against it                                                                                                                                                                                                                                                                                                                                                                                           | Required data (exact)                                                                                                                                                                                                            | Required market/<br>instrument access (exact)                                                                                                                             | Fit with current                                                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| C order-flow /<br>order-book /<br>microstructure | <b>Information-rich microstructure state:</b> order book imbalance / order flow features predict short-horizon returns; liquidity provision / price discovery dynamics. | Coinbase-specific study finds order book state impacts price discovery and limit orders (due to abundance) contribute materially. <sup>18</sup> Order-flow paper (JFM 2026) reports order flow predicts crypto returns with economic value, robust portfolio sorts. <sup>19</sup> Coinbase WebSocket <b>level2</b> feed provides snapshot/update semantics (new_quantity is absolute, event_time included) intended for building order books. <sup>20</sup> | Implementability is the enemy: microstructure requires correct L2 state + sequencing + high-volume storage + replay; otherwise “alpha” is often feed artifact. Coinbase WS rate limits are 8 req/sec per IP, and channels close without heartbeats. <sup>21</sup> Most academic “order flow” predictors are not simply “public L2,” but require comprehensive cross-venue flow or institutional datasets. <sup>19</sup> | Tick/trade prints + full L2 snapshots + all updates, timestamped; storage of raw updates for replay; execution-at-spread simulation. Coinbase channel semantics specify snapshot/update events and absolute sizes. <sup>20</sup> | Spot is enough for many microstructure strategies, but you need real-time WebSocket ingestion and (ideally) colocated/low-latency execution to compete at short horizons. | <b>Poor fit now:</b> rebar-based paper-n no even store/re engine. Analytic Review position microst as later work. (Analytic Review pp.3–4, |

| Lane                                                | Mechanism (edge hypothesis)                                                                                                                                      | Best supporting evidence                                                                                                                                            | Strongest evidence against it                                                                                                                                                                                                                                                                                                                                                              | Required data (exact)                                                                                                                          | Required market/ instrument access (exact)                                                                   | Fit with current                                                                                                                  |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>D</b> BTC-only indicator mean reversion fallback | <b>Intraday/ short-horizon reversal</b> in BTC: fade extremes or micro mean reversion using bar-level indicators; generally long-only or “cash vs long” in spot. | Intraday crypto paper finds evidence of both momentum and reversal patterns (including BTC) and reports timing strategy economic value vs benchmarks. <sup>22</sup> | Large crypto technical-rule study reports <b>no BTC OOS predictability</b> while other cryptos do — directly undermines BTC-only indicator hunting. <sup>6</sup> Your own repo hard-codes a BTC long-only breakout state machine and your <code>research_plan.md</code> documents that BTC single-strategy search produced too few trades / unstable inference (trade count failure mode). | BTC-USD OHLCV at 5m/ 1h/4h; fees + slippage; regime labels if using regime gating. Candle endpoint supports those granularities. <sup>11</sup> | Spot BTC only (if long/cash). If true long-short is needed, you’re back in derivatives (Lane B), not Lane D. | Feasible to implement quickly, evidence mixed/ negative, robust only in edge. A repeats “single-symbol” fragility plan was about. |

## Primary lane and backup lane

### Primary: Lane A (with a strict “portfolio form or it doesn’t count” rule)

Lane A is the best next move **because it addresses both sides of the problem**:

- 1) **Edge hypothesis has some support in crypto when pursued as a large-library + multiple-testing-controlled + portfolio construction program**, not as one BTC rule. <sup>1</sup>
- 2) **It directly fixes the structural trap you’re in**: the repo and the prior process are single-symbol BTC; the most damaging weakness is low effective sample size + selection bias. Lane A is how you raise sample size (more assets, more independent opportunities) and make research standards enforceable. <sup>24</sup>
- 3) **It is implementable on spot without requiring a full derivatives risk architecture (Lane B) or an L2 replay engine (Lane C)**. That’s a feasibility point — but feasibility alone is *not* the reason it’s first. <sup>25</sup>

What Lane A is **not**: “grab BTC candles and test 200 RSI variants.” That is exactly the failure mode White (2000) warns about. <sup>9</sup>

## Backup: Lane D (BTC mean reversion) as a fast baseline + sanity check

Lane D is the backup because it is the **cheapest to falsify** once you have even a minimal bar-based backtest harness. But the evidence base for a durable BTC-only indicator edge is mixed-to-negative, including explicit BTC OOS null results in large technical-rule literature. <sup>26</sup>

## Why the other two lanes are not first

### Why Lane B (carry/basis/funding) is not first:

Mechanism is strong, but *current implementability* is poor and risk is structurally different. Coinbase perps are gated by region eligibility and onboarding, require margin collateral, and impose liquidation risk where losses can exceed margin/collateral. <sup>27</sup> BIS also shows carry is largest in booms and predicts crashes; you don't get “clean yield,” you inherit crash/liquidation tail risk. <sup>12</sup> In repo terms, Lane B implies a **separate margin/liquidation-aware architecture**, not just a new signal.

### Why Lane C (microstructure) is not first:

Edge plausibility exists (order book state affects price discovery; order flow predicts returns), but you currently lack the binding capabilities: L2 state, sequencing validation, event storage, replay simulation, and execution-at-spread modeling. <sup>28</sup> Without those, you cannot falsify Lane C honestly in two weeks — you'd be “discovering” feed artifacts. That's a downgrade based on implementability, not a claim that microstructure is worthless. <sup>20</sup>

## Why the prior single-symbol BTC search process was structurally weak

This was not “bad indicator choice.” It was a structural research design problem:

- **Single-history + repeated reuse + parameter tweaks = selection bias by construction.** White (2000) exists specifically because with a single time series, repeated model selection produces false positives unless corrected. <sup>9</sup>
- **BTC-only means low independent opportunity count.** Your own breakout state machine shows a multi-stage funnel that can easily collapse to near-zero trades because each additional gate multiplies selectivity. Low trade count makes results statistically unstable — your `research_plan.md` explicitly quotes a “minimum of 300 independent trades” standard (even though its text is abbreviated with ellipses).
- **Costs were not first-class, and paper mode bakes in unrealistic assumptions.** `PaperAdapter` sets `commission: 0.0`, and slippage is inconsistent across config vs risk runtime. Anghel (2022) shows that omitting fees or liquidity costs can inflate false discoveries even under conservative multiple testing. <sup>5</sup>
- **The repo currently couples “strategy logic” to runtime control flow.** The `StateMachine` encodes the breakout strategy directly, so iterating strategies becomes “edit production logic,” which encourages premature optimization and discourages controlled research runs. <sup>3</sup>



Lane A is chosen largely because it is the cleanest structural antidote: **increase breadth, pre-register the search, enforce multiple-testing + walk-forward, then only consider portfolio-form survivors.** <sup>29</sup>

## Falsification tests and two-week kill criteria

### E. Cheapest falsification test for each lane (doable within 2 weeks)

#### Lane A (OHLCV multi-asset portfolios):

Cheapest falsification = build the *minimum viable research harness* and run a **small, pre-registered** library across a **multi-asset universe** (e.g., top 15–30 spot pairs by volume proxy), with strict walk-forward and a multiple-testing control step.

- Why this is the cheapest honest test: the core risk is false discovery; you must include the defenses in the falsification, not after. <sup>23</sup>

- Data: candles via Coinbase product candles endpoint, paginated. <sup>11</sup>

#### Lane B (carry/basis/funding):

Cheapest falsification = **eligibility + data feasibility check first**, then a paper backtest of a basic carry variant with conservative assumptions (fees, funding settlement timing, stress around spike regimes).

- If you cannot access perps (region/onboarding), the lane is killed before research begins. <sup>30</sup>

- If you can, use Coinbase funding definitions + liquidation rules to model tail risk realistically. <sup>31</sup>

#### Lane C (microstructure):

Cheapest falsification = record **2–5 trading days** of `level2` + `market_trades` for 1–2 symbols, build book state, compute a tiny feature set (book imbalance, order flow imbalance proxies), and test whether predicted next-horizon returns exceed half-spread + fees.

- Coinbase provides the channel semantics needed to build a snapshot + updates book and has explicit rate limits (8/sec per IP). <sup>32</sup>

- This is still non-trivial; that's exactly why Lane C is downgraded on implementability.

#### Lane D (BTC-only mean reversion):

Cheapest falsification = implement **one** pre-specified BTC mean reversion rule (not 30 variants) + strict walk-forward (multiple windows) + realistic costs; compare to buy-and-hold and to “cash-only.”

- Intraday predictability literature supports that reversal can exist, but it is regime/condition dependent. <sup>22</sup>

- BTC OOS technical-rule null results and prior failures are the negative prior. <sup>6</sup>

### H. Two-week kill criteria (explicit per lane)

**Lane A — kill within 2 weeks if any of these happens:** 1) You cannot assemble a usable multi-asset OHLCV dataset from Coinbase candles with sufficient history to run at least **3 walk-forward windows** (data feasibility hard-stop). <sup>11</sup>

2) After including realistic friction (fees + slippage) and applying multiple-testing control, **no** rule family produces a portfolio/ensemble that is net-positive out-of-sample and not hypersensitive to small parameter perturbations (classic overfit signature). <sup>33</sup>

3) Survivors only exist because of unrealistic paper assumptions (e.g., zero commission), meaning the lane is “artefact-positive” not “market-positive.”

**Lane B — kill within 2 weeks if any of these happens:** 1) You are not eligible / cannot onboard to perps (hard gate). <sup>30</sup>

2) Conservative simulation shows returns are dominated by liquidation/crash tails (carry spikes coincide with crash risk) or require leverage levels that create unacceptable liquidation risk under Coinbase's own liquidation disclosures. <sup>34</sup>

3) You cannot obtain or reliably reconstruct funding/basis histories at correct settlement timing, making falsification impossible. <sup>16</sup>

**Lane C — kill within 2 weeks if any of these happens:** 1) You cannot maintain a consistent L2 book (sequence gaps/drift; missing updates), or operational constraints (rate limits / channel closures without heartbeats) prevent stable capture. <sup>32</sup>

2) Any apparent "predictability" fails immediately once you simulate crossing spread + fees (i.e., predicted move < friction). <sup>20</sup>

3) You cannot build a replayable event-store quickly enough to separate signal from implementation artifacts (honest kill: engineering insufficiency). <sup>20</sup>

**Lane D — kill within 2 weeks if any of these happens:** 1) OOS expectancy is  $\leq 0$  after friction, or results flip sign with small parameter changes (overfit). <sup>35</sup>

2) Performance is confined to one regime and collapses across walk-forward windows (non-stationarity). <sup>22</sup>

3) The rule requires shorting to be viable; at that point it's not Lane D anymore — it's a derivatives lane (B). <sup>30</sup>

## Engineering tasks for chosen lane

### F. Exact 7 engineering tasks for Lane A (and only what's needed to falsify it)

1) **Unify risk + slippage to be config-driven:** make `bot/risk.py` read `risk_per_trade` and `max_entry_slippage_bps` from `config.yaml` so research runs aren't invalidated by hidden constants.

2) **Add realistic cost hooks to paper execution** (even if not perfect): stop returning `commission: 0.0` unconditionally in paper fills (or inject fee/slippage at the backtest layer).

3) **Implement a Coinbase candle downloader + paginator** for a list of spot products and time ranges (max 350 candles/request). <sup>11</sup>

4) **Create a local dataset cache + manifest** (parquet/CSV + metadata: symbol, timeframe, start/end coverage, missing bars).

5) **Build a minimal bar-based backtest engine** that can run strategies on OHLCV without lookahead and with friction. (This is explicitly called out as missing in the Analytical Review PDF.)

6) **Implement walk-forward runner** (rolling windows, train-select → test-evaluate) compatible with parameter grids. <sup>9</sup>

7) **Implement a minimal multiple-testing control step** suitable for rule libraries (start with FDR-style selection or a reality-check style bootstrap) so you can distinguish edge from chance. <sup>36</sup>

# Research tasks for chosen lane

## G. Exact 7 research tasks for Lane A (pre-registered, portfolio-form)

- 1) **Define the trading universe** (spot-only): select 15–30 Coinbase spot pairs (objective filter: availability + liquidity proxy), and freeze the list for the first experiment. <sup>11</sup>
- 2) **Pre-register the rule families + parameter bounds** and cap the total number of tested configurations (explicit defense against “search until it works”). <sup>9</sup>
- 3) **Pre-register the evaluation design**: walk-forward window sizes (e.g., 180d train / 60d test, step 60d), metrics, and cost model. <sup>23</sup>
- 4) **Run a pilot (small) study**: 5 assets × 2 timeframes × 2 rule families to validate no leakage and reasonable trade frequency.
- 5) **Run the full library search** across the full universe; apply the multiple-testing control step and keep only survivors. <sup>37</sup>
- 6) **Construct portfolios/ensembles of survivors** (equal-weight or volatility-scaled) and evaluate OOS stability; do not promote single-rule BTC winners. <sup>38</sup>
- 7) **Write the falsification report**: number of trials, number of survivors after costs, OOS performance distribution, and sensitivity to friction/parameters (explicitly documenting whether results are robust or fragile). <sup>35</sup>

## Final recommendation

### I. Build now, pretest only, or stop and reassess

**Recommendation: Build now** — but “build” means **build the minimal Lane A research harness** (data → backtest → walk-forward → multiple-testing → portfolio evaluation), not “add more runtime logic.” This is the cheapest path to a credible yes/no on the best spot-compatible lane, and it directly addresses the structural failure mode of single-symbol BTC searching. <sup>39</sup>

If Lane A fails its two-week kill rules, the correct next action is **stop and reassess the entire OHLCV-only premise** (and potentially reconsider Lane B only if derivatives access + risk infrastructure are realistically achievable), rather than sliding back into BTC-only indicator iteration. <sup>40</sup>

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<sup>1</sup> <sup>6</sup> <sup>26</sup> Technical trading and cryptocurrencies | Annals of Operations Research | Springer Nature Link  
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